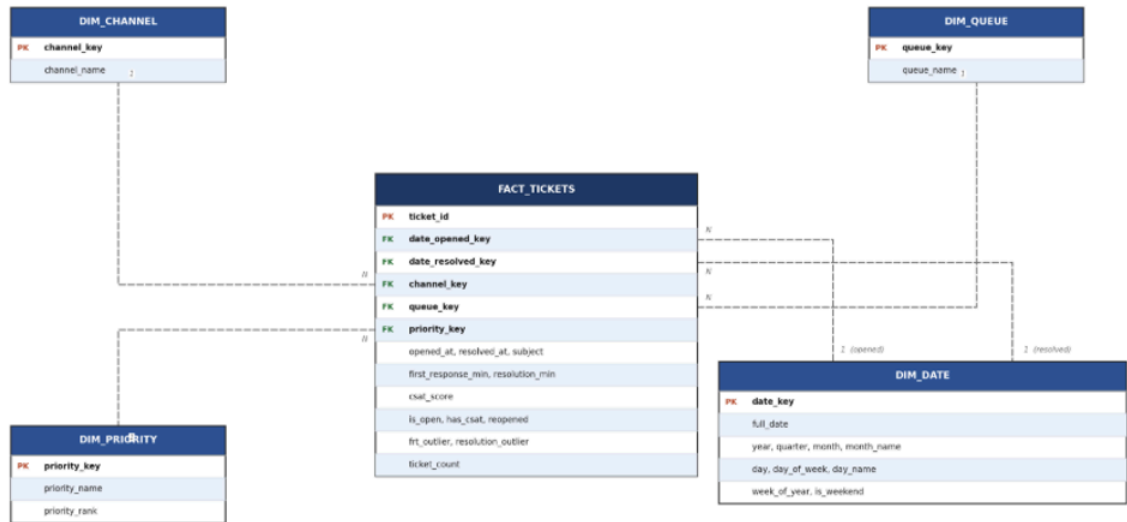


CUSTOMER SUPPORT TICKET ANALYTICS

Data Warehouse Design

1. Dimensional Model



The dimensional model follows a star schema with **FACT_TICKETS** at the center and four supporting dimensions: **DIM_DATE**, **DIM_CHANNEL**, **DIM_QUEUE**, and **DIM_PRIORITY**.

The declared grain of **FACT_TICKETS** is one row per support ticket. **DIM_DATE** is a role-playing dimension used through two foreign keys: **opened_date_key** and **resolved_date_key**.

Table	Type	Grain / Contents
FACT_TICKETS	Fact	One row per support ticket; contains dimension foreign keys, operational measures, and analytical flags
DIM_DATE	Dimension	One row per calendar day; role-playing dimension for opened and resolved dates
DIM_CHANNEL	Dimension	One row per support channel
DIM_QUEUE	Dimension	One row per support queue
DIM_PRIORITY	Dimension	One row per priority level

2. PostgreSQL Table Definitions

2.1 DIM_DATE

```
CREATE TABLE dim_date (  
    date_key      INT      NOT NULL, -- surrogate key, format YYYYMMDD  
    full_date     DATE      NOT NULL,  
    year          SMALLINT NOT NULL,  
    quarter       SMALLINT NOT NULL,
```

```

month          SMALLINT  NOT NULL,
month_name     VARCHAR(15) NOT NULL,
day            SMALLINT  NOT NULL,
day_of_week    SMALLINT  NOT NULL, -- 1 = Monday ... 7 = Sunday
day_name       VARCHAR(15) NOT NULL,
week_of_year   SMALLINT  NOT NULL,
is_weekend     BOOLEAN   NOT NULL,
CONSTRAINT pk_dim_date PRIMARY KEY (date_key)
);

```

2.2 DIM_CHANNEL

```

CREATE TABLE dim_channel (
    channel_key  INT          NOT NULL, -- surrogate key
    channel_name VARCHAR(20) NOT NULL, -- chat, email, in-app, phone
    CONSTRAINT pk_dim_channel PRIMARY KEY (channel_key),
    CONSTRAINT uq_dim_channel_name UNIQUE (channel_name)
);

```

2.3 DIM_QUEUE

```

CREATE TABLE dim_queue (
    queue_key  INT          NOT NULL, -- surrogate key
    queue_name VARCHAR(30) NOT NULL, -- account, billing, integrations,
                                     -- onboarding, technical
    CONSTRAINT pk_dim_queue PRIMARY KEY (queue_key),
    CONSTRAINT uq_dim_queue_name UNIQUE (queue_name)
);

```

2.4 DIM_PRIORITY

```

CREATE TABLE dim_priority (
    priority_key  INT          NOT NULL, -- surrogate key
    priority_name VARCHAR(10) NOT NULL, -- low, normal, high, urgent
    priority_rank SMALLINT    NOT NULL, -- 1=low ... 4=urgent, for sorting
    CONSTRAINT pk_dim_priority PRIMARY KEY (priority_key),
    CONSTRAINT uq_dim_priority_name UNIQUE (priority_name)
);

```

2.5 FACT_TICKETS

```

CREATE TABLE fact_tickets (
    ticket_id      VARCHAR(15) NOT NULL, -- degenerate dimension / PK
    date_opened_key INT          NOT NULL, -- FK -> dim_date
    date_resolved_key INT        NULL,    -- FK -> dim_date; NULL = still
open
    channel_key     INT          NOT NULL, -- FK -> dim_channel
    queue_key       INT          NOT NULL, -- FK -> dim_queue
    priority_key     INT          NOT NULL, -- FK -> dim_priority

```

```

        opened_at          TIMESTAMP NOT NULL, -- full timestamp, for time-of-day
analysis
        resolved_at        TIMESTAMP NULL,
        subject            VARCHAR(255) NOT NULL, -- free text; drill-through only, not
aggregated
        first_response_min NUMERIC(10,2) NULL,      -- measure; NULL = never
responded
        resolution_min     NUMERIC(12,2) NULL,      -- measure; NULL until resolved
        csat_score         SMALLINT NULL,          -- measure; 1-5, NULL = not surveyed
        is_open            BOOLEAN NOT NULL, -- flag: resolved_at IS NULL
        has_csat           BOOLEAN NOT NULL, -- flag: csat_score IS NOT NULL
        reopened           BOOLEAN NOT NULL, -- flag
        firt_outlier       BOOLEAN NOT NULL, -- flag: first_response_min beyond IQR
fence
        resolution_outlier BOOLEAN NOT NULL, -- flag: resolution_min beyond IQR
fence
        ticket_count       SMALLINT NOT NULL DEFAULT 1, -- additive helper
measure
        CONSTRAINT pk_fact_tickets PRIMARY KEY (ticket_id),
        CONSTRAINT fk_fact_date_opened FOREIGN KEY (date_opened_key)
REFERENCES dim_date (date_key),
        CONSTRAINT fk_fact_date_resolved FOREIGN KEY (date_resolved_key)
REFERENCES dim_date (date_key),
        CONSTRAINT fk_fact_channel FOREIGN KEY (channel_key) REFERENCES
dim_channel (channel_key),
        CONSTRAINT fk_fact_queue FOREIGN KEY (queue_key) REFERENCES
dim_queue (queue_key),
        CONSTRAINT fk_fact_priority FOREIGN KEY (priority_key) REFERENCES
dim_priority (priority_key),
        CONSTRAINT chk_csat_range CHECK (csat_score IS NULL OR csat_score
BETWEEN 1 AND 5)
);

```

4. Dashboard Indexes

Helpful indexes for the dashboard's most common filters and group-bys.

```

CREATE INDEX idx_fact_date_opened ON fact_tickets (date_opened_key);
CREATE INDEX idx_fact_date_resolved ON fact_tickets (date_resolved_key);
CREATE INDEX idx_fact_channel ON fact_tickets (channel_key);
CREATE INDEX idx_fact_queue ON fact_tickets (queue_key);
CREATE INDEX idx_fact_priority ON fact_tickets (priority_key);
CREATE INDEX idx_fact_is_open

```